

SONIYA KODIPUNJULA

Dallas, Texas | +1-940-222-1165 | soniyakodipunjula16997@gmail.com | [Linkedin](#)

PROFILE

SAP Security and GRC Consultant with 4.7 years of progressive experience in implementing and supporting **SAP S/4HANA, Fiori (Spaces & Pages)**, and **SAP GRC Access Control modules (ARA, ARM, BRM, EAM)**. Proven expertise in **SAP Identity Management (IDM)** for centralized user provisioning and **SAP Solution Manager (SolMan)** for change and transport management. Proficient in configuring **BRF+**, **MSMP workflows**, and conducting **Segregation of Duties (SoD)** risk analysis. Experienced in leading **SAP Security audits, system upgrades**, and managing end-to-end authorization models across **BW/4HANA, SAP Analytics Cloud (SAC)**, and **Master Data Governance (MDG)**. Strong knowledge of **regulatory compliance frameworks, SOX controls**, and **security hardening best practices**.

PROFESSIONAL EXPERIENCE

Deloitte – Advisory Associate Solution Advisor

Dec 2021 – Jul 2023

Environments: SAP GRC 12.0 (Access Control, EAM), IAM, SAP FIORI, SAP S/4HANA, SAP Cloud, SAP BW

Client Projects: AMD & Dell, NC

- Designed and implemented role-based security concepts for **SAP S/4HANA** and **SAP Fiori** applications, integrating with enterprise risk and compliance standards in a complete life-cycle implementation project.
- Configured and supported all key **SAP GRC Access Control** modules:
 - **Access Risk Analysis (ARA)** – Performed SoD risk analysis and mitigation strategies.
 - **Access Request Management (ARM)** – Implemented BRF+ routing rules and MSMP workflow approvals.
 - **Emergency Access Management (EAM)** – Controlled firefighter ID usage with audit compliance.
 - **Business Role Management (BRM)** – Built role libraries and managed role lifecycle.
 - **User Access Review (UAR)** – Conducted periodic reviews to maintain access hygiene.
- Leveraged **SAP IDM** for centralized user provisioning, synchronization across SAP landscapes, and integration with GRC workflows.
- Used **SAP Solution Manager (SolMan)** to manage change requests, analyze transport impact, and enforce secure deployment practices.
- Conducted SAP Security audits, reviewed **vulnerability notes**, and remediated critical authorizations and objects.
- Designed and implemented **BW/4HANA, SAP Analytics Cloud (SAC)**, and **SAP MDG** security models, including object-level data authorizations.
- Led SAP Security and GRC upgrades across landscapes, ensuring business continuity and regulatory compliance.
- Coordinated with internal stakeholders, audit teams, and external vendors to drive implementation projects to successful closure.
- Provided guidance to junior security team members and contributed to security design reviews and documentation.

Cognizant Technology Solutions – Programmer Analyst

Jan 2019 – Nov 2021

Environments: SAP R/3, ECC 6.0, SAP GRC 10.0 (Access Control, EAM), IDM, SAP FIORI, SAP S/4HANA

Client Project: SANOFI AVENTIS, NA

- Efficient SAP Security Consultant with hands-on experience in **ECC, S/4HANA, GRC Access Control (ARA, ARM, EAM)**, and **IDM** modules, supporting both **incident and change management** processes.
- Performed comprehensive **user maintenance and role administration**, including the design, creation, and modification of **single, derived, and composite roles**; maintained user groups, profiles, and ensured alignment with **SOX and SoD compliance** requirements.
- Managed **transport requests** using **ChaRM in SAP Solution Manager (SolMan)**; resolved SolMan-specific issues and coordinated with BASIS on troubleshooting short dumps and RFC-related errors.
- Conducted **authorization analysis** using **SU53, ST01, and SUIM** to resolve missing authorizations and critical access issues; ran **user-level and role-level simulations** to proactively detect conflicts.
- Supported full lifecycle of **SAP GRC Access Control** processes:
 - **ARA** – Performed SoD risk analysis, critical transaction monitoring, and simulations.
 - **ARM** – Monitored access request statuses, resolved workflow issues, and managed BULK user provisioning.

- **EAM** – Created firefighter IDs, provisioned emergency access, maintained FFID owners, and executed monthly FF log reviews and audit trail validations.
 - Conducted **role uploads**, updated role approvers, and performed deactivation activities as needed.
 - Collaborated with **Internal Controls and Audit (PwC)** to provide audit evidence, maintain compliance reports, and implement **mitigating controls** based on evolving risk IDs and SoD rule updates.
 - Administered **SAP IDM** operations, including password parameter updates, sync troubleshooting, and health monitoring.
 - Provisioned and troubleshot **SAP Portal access**, assigned portal roles, and resolved GUI-related authorization issues.
 - Supported **critical incident handling**, generated audit and activity reports, and ensured secure and traceable access management across systems.
 - Prepared detailed **SOPs, knowledge base articles**, and mentored junior team members; actively proposed **continuous service improvement (CSI)** initiatives to promote client satisfaction and operational excellence.
 - Delivered **best-in-class client services**, consistently maintaining secure authorization landscapes and protecting sensitive business data.
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ACADEMIC PROJECT EXPERIENCE

Eventique – AI-Enhanced Event Planning Web Application | Software Engineering

GitHub: [Eventique/eventique-main at main · SoniyaKodipunjula/Eventique](https://github.com/SoniyaKodipunjula/Eventique)

Group Project | Role: Team Member

Tech Stack: React.js, Node.js, Python, MongoDB, Nginx, GCP, Docker, GitHub Actions, Jira

- Developed a responsive AI-powered event planning platform with smart recommendations, scheduling, and notifications.
 - Implemented features like calendar task editing and instant plan creation to enhance user experience.
 - Collaborated in **Agile** sprints using **Jira** for task tracking and sprint planning.
 - Resolved frontend-backend integration issues and improved UI consistency.
 - Set up **CI/CD pipelines with Docker and GitHub Actions**; deployed via **Google Cloud Run**.
 - Monitored app performance through **logging** and **DevOps metrics**; contributed to sprint documentation and user stories.
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SecureAI – Real-Time Phishing & Malware Detection Web Application | Secure E-commerce

GitHub: github.com/SoniyaKodipunjula/SecureAIWeb

Group Project | Role: Team Member

Tech Stack: Flask, Python, scikit-learn, Joblib, Flask-Talisman, OWASP ZAP, Render, HTML/CSS, Bootstrap, Git

- Developed a secure e-commerce platform to detect phishing URLs and malware-infected files using trained ML models.
 - Implemented AI-driven backend for real-time classification and hash-based malware detection.
 - Integrated HTTPS, CSRF protection, and input validation using Flask-Talisman for end-to-end security.
 - Scanned vulnerabilities using OWASP ZAP, Nessus, and Nikto, addressing SQLi, XSS, and DoS threats.
 - Deployed to Render using CI/CD pipeline with environment hardening via render.yaml.
 - Designed login/MFA flow and user-friendly UI using Bootstrap, collaborating across all architecture layers.
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Survival Analysis Using Logistic Regression | Introduction to Big Data and Data Science

Group Project | Role: Statistical Analyst

Tech Stack: Python, Jupyter, Pandas, Scikit-learn, Matplotlib

- Performed statistical analysis (t-tests, chi-square) on a large healthcare dataset (~149K records) comparing Appalachia and non-Appalachia regions.
 - Identified key predictors of survival based on age, race, surgical treatment, and location.
 - Developed a logistic regression model to predict survival outcomes with 60%+ accuracy.
 - Collaborated using Git and Jupyter for version control and reproducible workflows.
 - Gained experience in EDA, hypothesis testing, and data-driven healthcare insights.
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Research Project – COVID-19 Spread Forecasting Using GNNs | Graph Theory

Group Project | Role: Co-Developer

Tech Stack: PyTorch, DGL, NumPy, Pandas, Matplotlib

- Developed predictive models combining SIR epidemiological simulations with Graph Neural Networks (GNNs) to forecast COVID-19 spread in Italy.
- Built wide and deep GNN architectures capturing mobility and geographic relationships across provinces.
- Used MSLE for model evaluation and compared GNN, SIR, and hybrid performance.
- Handled data normalization, graph construction, and modeling policy-change impact scenarios.
- Addressed limitations such as data bias, assumption drift, and external confounders.

Agent-Based Models & Neural Networks | Natural Language Processing

Role: Individual Contributor

Tech Stack: PyTorch, Pandas, NumPy

- Modified Agent-Based Models with Bayesian inference to estimate linguistic dynamics like variant prevalence, migration time, and imbalance.
- Replaced spline functions with sigmoid-based imbalance modeling, assigning traits at agent birth.
- Built two PyTorch neural network models to classify language families: one using phonological feature vectors, another using learned embeddings.
- Trained a CBOW-based word generator on orthographic English words and generated valid outputs.
- Evaluated using Matthews Correlation Coefficient (MCC) and documented modeling decisions thoroughly.

EDUCATION

University of North Texas – Denton, Texas, United States

Master of Science | Computer Science | GPA: 3.5 | Aug 2023 – May 2025

Relevant Coursework:

Natural Language Processing, Secure E-Commerce, Information Retrieval, Machine Learning, Big Data & Data Science, Graph Theory, Software Engineering, Data Modeling, Intro to Computer Security, Computer Networks, Computer Architecture

Sri Venkateswara College of Engineering & Technology – Chittoor, Andhra Pradesh, India

Bachelor of Technology | Electrical & Electronics Engineering | GPA: 3.5 | Sep 2014 – Apr 2018

TECHNICAL SKILLS

- **SAP Systems:** SAP R/3, ECC 6.0, S/4HANA, BW/4HANA, SAP Fiori, SAP Analytics Cloud (SAC), SAP GRC Access Control (ARA, ARM, BRM, EAM)
- **SAP Security Administration:** User Administration, Role Design & Implementation, Role Maintenance, Authorization Object Management
- **GRC & Compliance:** SoD Management, Risk Mitigation & Remediation, Incident Response, Risk Reporting, Audit Support
- **Cloud Security:** Experience with AWS, Microsoft Azure, and SAP Cloud Platform environments
- **Security Standards & Compliance:** SOX, GDPR, ITGC, Internal Controls, PwC Audit Coordination
- **Testing & Validation:** Security Unit Testing, Functional Testing, UAT (User Acceptance Testing)
- **Programming & Automation:** Python, Java, SQL; CI/CD pipeline experience using GitHub Actions, Docker, Jenkins
- **IT Service Tools:** ServiceNow (Incident/Change Management), WinSCP, qTest, Citrix
- **Documentation & Reporting:** SOP Development, Testing Reports, Audit Logs, Firefighter Logs, Transport Logs, Continuous Service Improvement (CSI) Proposals